VINAY KUMAR REDDY SIRIGIREDDY

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PROFESSIONAL SUMMARY

M.S. Aerospace Systems Engineering (Wright State University, GPA 3.7, May 2026) with hands-on research experience executing FHA, FMEA, and fault propagation analysis on safety-critical eVTOL propulsion architectures per ARP 4761A. Developed a dual-constraint MBSE sizing framework in CATIA Magic/SysML and MATLAB that predicted MTOW within 1.5% across three certified industry platforms — Wisk Cora, Joby S4, and BETA ALIA-250 — using published specs only, with no vehicle-specific tuning. Authored two journal manuscripts currently under peer review targeting Advanced Air Mobility applications. Proficient across the full systems engineering lifecycle: requirements decomposition, architecture design, hazard analysis, automated V&V, and digital thread maintenance.

CORE COMPETENCIES

Safety Engineering & Standards: ARP 4761A (FHA, FMEA, FTA, CCA/CMA), ARP 4754A, FAA 14 CFR Part 23/27, MIL-STD-882 (awareness), DO-178C/DO-254 (awareness), hazard identification, risk quantification, mitigation verification

MBSE & Systems Engineering: SysML (BDD, IBD, PAR, REQ, STM, ACT, UC), CATIA Magic / Cameo Systems Modeler, requirements traceability, architecture decomposition, interface definition, V&V planning and execution, digital thread

Analysis & Simulation: MATLAB/Simulink, OpenVSP, ANSYS (FEA), Cradle CFD, HyperMesh, NASA CEA, AGI STK, XFOIL, Python

CAD & Design: SOLIDWORKS (CSWP, CSWA), CATIA V5, AutoCAD, Fusion 360; GD&T, engineering tolerancing, drawing standards

Programming & Tools: MATLAB, Python (data science, scripting, automation), JavaScript / React (full-stack web tooling)

Aerospace Domains: eVTOL / UAS safety-critical systems, Distributed Electric Propulsion, Advanced Air Mobility (AAM/UAM), Aircraft Performance, Aerodynamics, Propulsion, Structural Analysis

TECHNICAL EXPERIENCE

Graduate Research Assistant — Wright State University, Dayton, OH *Feb 2025 – May 2026*

Advisor: Dr. Darryl K. Ahner, Dean, College of Engineering & Computer Science

- **Reduced MTOW prediction error to <1.6% across three industry eVTOL platforms** (Wisk Cora: 0.98%, Joby S4: 1.28%, BETA ALIA-250: 1.51%) by developing a dual-constraint sizing framework that simultaneously enforces peak hover power and total mission energy limits — eliminating the 72.4% MTOW underestimation produced by energy-only methods.
- **Identified energy-only battery sizing as a Category I safety hazard** through applied FHA: quantified a 1,247 kg MTOW underestimate that rendered hover thrust margins physically infeasible, then implemented dual-constraint reformulation to resolve the failure mode — a direct application of ARP 4761A hazard severity assessment and design-change mitigation.
- **Automated airworthiness compliance verification for 17 design configurations** by encoding FAA 14 CFR Part 23 Amendment 64 constraints as executable Boolean expressions inside SysML parametric constraint blocks bound to MATLAB/Simulink co-simulation — eliminating manual pass/fail interpretation and maintaining a fully auditable compliance record.
- **Built a 10-diagram SysML fault-propagation architecture (BDD, IBD, STM, ACT, PAR, REQ, UC)** for a hybrid Lift+Cruise / Tilt-Rotor eVTOL, tracing single-point failure propagation across motor, inverter, and battery subsystems and verifying containment boundaries per ARP 4761A FHA and FMEA processes.
- **Maintained full requirements traceability across the MBSE digital thread** by decomposing safety-relevant requirements into a SysML Requirements Diagram with unique IDs linked to downstream design parameters, ensuring hazard mitigations remain traceable to originating requirements across all 17 evaluated configurations.

Advanced Rocket Propulsion Intern — MARS Exploration Pvt. Ltd., India *May – Aug 2022*

- **Improved predicted combustion efficiency by ~12%** for hybrid RP-1/Al + LOX propulsion concepts by sweeping O/F ratios from 1.0 to 3.0 in NASA CEA, isolating the 1.5–2.0 range as optimal for specific impulse (Isp) and identifying mixture ratios that degraded combustion performance — directly informing propellant selection for ion thruster trade studies.

STK Mission Analysis Intern — SS Technologies, Bengaluru, India *Nov 2021 – Feb 2022*

- **Evaluated mission performance under environmental and operational constraints** by modeling satellite orbital mechanics and multi-aircraft trajectory profiles in AGI STK, verifying ground-track coverage and communications link geometry against mission requirements.

CAD Design Intern — Verzeo, Bengaluru, India *Aug – Oct 2021*

- **Produced a library of dimensioned 2D and 3D mechanical components in AutoCAD** to engineering drawing standards, developing proficiency in GD&T, tolerance stack-up analysis, and production drawing formats — skills directly applied to subsequent FEA and structural optimization work.

SELECTED PROJECTS

SysML Fault-Propagation Architecture — Hybrid LPC/TR eVTOL | CATIA Magic / Cameo 2024 – Present

- Constructed a 10-diagram SysML model tracing failure propagation across propulsion, battery, and flight control subsystems. Defined failure containment boundaries and interface safety requirements using ARP 4761A FHA and FMEA; state machine diagrams capture transition logic for fault-induced mode changes.

eVTOL Aircraft Sizing Web Application | Systems Architect & Technical Director | evtolsizer.live 2024 – 2025

- Defined all system requirements, governing physics models, and V&V acceptance criteria for a live 23-module sizing platform (evtolsizer.live): authored the hover/cruise power equations, FAA 14 CFR Part 23 constraint logic, ISA-corrected aeroacoustic propagation model, and Monte Carlo trade study architecture. Full-stack implementation was executed via AI-assisted code synthesis under direct technical direction; algorithm correctness and regulatory compliance were verified by the author throughout. Validated against Wisk Cora, Joby S4, and BETA ALIA-250 within 1.5% MTOW error — a research deliverable under Dr. Darryl K. Ahner.

CFD Analysis — Hoverbike Overlapping Propeller Configurations | Cradle CFD / SOLIDWORKS 2023

- Quantified aerodynamic interference losses and thrust-to-power tradeoffs for overlapping rotor layouts by running steady-state CFD simulations, informing rotor spacing recommendations for AAM propulsion system sizing.

FEA Structural Optimization — Hoverbike Duct Assembly | HyperMesh / ANSYS 2023

- Met all required structural safety factors at reduced mass by comparing carbon fiber composite layups against 6-series aluminum alloys under representative flight load cases — reducing component mass while maintaining structural margins.

20-Seat Business Jet Conceptual Design | OpenVSP / SOLIDWORKS 2022 – 2023

- Executed full conceptual sizing cycle from initial weight estimation through configuration layout and preliminary structural sizing, benchmarking aerodynamic performance and mass properties against the Gulfstream G500.

EDUCATION

Master of Science, Aerospace Systems Engineering — Wright State University, Dayton, OH May 2026

GPA: 3.7 | Thesis (Accepted April 2026): MATLAB-MBSE Integrated Sizing Framework for eVTOL Aircraft: Evaluating Peak Power Demands on MTOW Convergence | Advisor: Dr. Darryl K. Ahner

Bachelor of Science, Aeronautical Engineering — Bharath Institute of Higher Education and Research, India May 2023

GPA: 3.5

CERTIFICATIONS & PUBLICATIONS

SOLIDWORKS: Certified SolidWorks Professional (CSWP), Drawing Tools Professional, Additive Manufacturing Associate, CSWA

Other Certifications: Dassault Systemes 3D Swym Associate; IBM Python for Data Science, AI & Development; MathWorks MATLAB; NPTEL Aircraft Performance & Manufacturing Process Technology I/II; Udemy Product Development & Systems Engineering

Publication: Design and Fabrication of Hoverbike — IJISRT (aerodynamic and propulsion analysis)

ADDITIONAL

Professional Membership: ASME

Languages: English (full professional), Telugu (native), Hindi (full professional), Tamil